

VAPOR

Value-Model-Based Reinforcement Learning
for Advanced Reasoning

60.4 on AIME 2024

Surpassing DAPO by 10+ points | No SFT data | Qwen2.5-32B base

60.4

Why Value Models? Finer Credit Assignment for Long CoT

Value-model-free methods (GRPO, DAPO) assign trajectory-level advantage uniformly across all tokens — limiting credit assignment precision for complex reasoning.

Value-Model-Free (GRPO / DAPO)

- Uniform advantage across all tokens
- Monte Carlo return estimation
- Higher variance per-token estimates
- No generalization across samples

Value-Model-Based (VAPO)

- Token-level credit assignment
- Traces each action's impact on returns
- Lower-variance estimates per token
- Generalizes across samples

Critical when subtle step-level errors cause catastrophic failures in long chain-of-thought reasoning.

Three Challenges Plaguing Value-Model-Based RL in Long-CoT

01

Value Model Bias

Long trajectories and bootstrapped learning amplify estimation bias in the value model, causing compounding errors over extended reasoning chains.

02

Heterogeneous Sequence Lengths

Responses vary dramatically in length, requiring different bias-variance tradeoffs — a single fixed GAE λ parameter cannot serve all sequences optimally.

03

Reward Signal Sparsity

Only the final <eos> token receives reward, making credit assignment across thousands of reasoning tokens extremely difficult.

VAPO Framework: Integrated Solutions

Value Model Bias

Value-Pretraining

Pre-trains value model before RL to reduce initial bias

Decoupled-GAE

Separates GAE computation for robust value estimation (from VC-PPO)

Heterogeneous Lengths

Length-Adaptive GAE

NOVEL

Dynamically adjusts λ based on response length (novel)

Token-level Loss

Normalizes loss at token level, not sequence level (from DAPO)

Reward Sparsity

Clip-Higher

Asymmetric clipping encourages exploration (from DAPO)

Positive Example LM Loss

Self-imitation learning on correct trajectories

Group-Sampling

Multiple responses per prompt for better baseline (from GRPO)

Each technique targets a specific challenge — VAPO integrates all seven for stable, high-performance reasoning RL.

Length-Adaptive GAE: Dynamic λ for Variable Response Lengths

Core Insight: A single fixed λ cannot serve both short and long responses — the optimal bias-variance tradeoff depends on sequence length.

Short Responses

Lower λ

- Fewer tokens → less bootstrapping error
- Prioritizes lower variance
- More stable gradient estimates



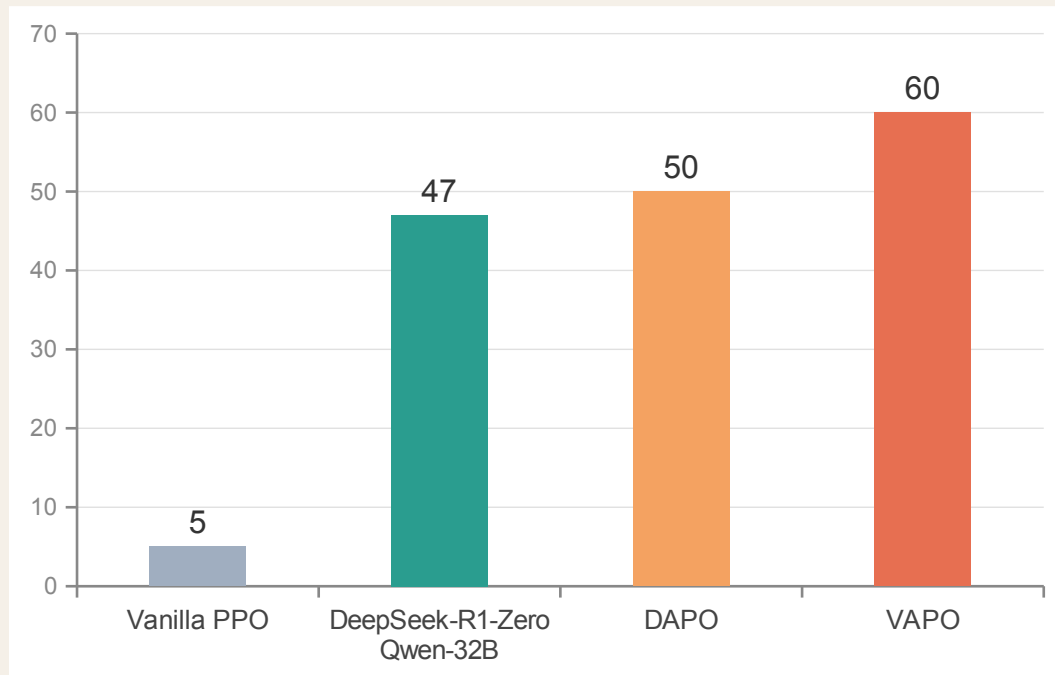
Long Responses

Higher λ

- More tokens → higher bootstrapping bias risk
- Prioritizes lower bias
- Better long-horizon credit assignment

Impact: Removing Length-Adaptive GAE drops AIME 2024 score from 60 to 45 (-15 points). This novel technique addresses the heterogeneous sequence length problem unique to long-CoT reasoning.

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO



Key Results

- All models use Qwen2.5-32B base with no SFT data
- VAPO surpasses DAPO by 10 points and DeepSeek-R1-Zero by 13 points
- Vanilla PPO scores only 5 — value-model-based RL fails without VAPO's innovations
- VAPO reaches 60.4 (avg@32) within ~5,000 training steps

See [figure_2.jpg](#) for training curves: VAPO consistently outperforms DAPO throughout training.

Training Efficiency: SOTA in 5K Steps with Zero Crashes

Training Curve — AIME 2024 Accuracy

(figure_2.jpg)



VAPO → 60% at step ~4,500



DAPO → 50% at step ~5,500



DeepSeek-R1-Zero → 47% at step ~10,000

VAPO consistently outperforms DAPO throughout the entire training run.

Response Length & Stability

(figure_3.jpg)

~6,000

tokens (VAPO)

~4,500

tokens (DAPO)

Zero training crashes across multiple independent runs. VAPO learns more elaborate reasoning chains while maintaining complete training stability.

Ablation: Every Component Matters — Value-Pretraining Adds 49 Points

Configuration	AIME 2024 Score	Δ
VAPO (full)	60	
w/o Value-Pretraining	11	-49
w/o Decoupled-GAE	33	-27
w/o Length-Adaptive GAE	45	-15
w/o Clip-Higher	46	-14
w/o Token-level Loss	53	-7
w/o Pos. Example LM Loss	54	-6

Critical: Removing Value-Pretraining drops score from 60 $\rightarrow$ 11. Without it, value-model-based RL barely works. The top three components (Value-Pretraining, Decoupled-GAE, Length-Adaptive GAE) collectively contribute 91% of the gain.

Key Takeaways

1

Value-model-based RL can surpass value-model-free methods for complex reasoning when the three core challenges (bias, heterogeneous lengths, reward sparsity) are systematically addressed.

2

Value-Pretraining is the single most impactful component, contributing 49 points (from 11 to 60). Without it, value-model-based RL barely functions.

3

Length-Adaptive GAE is a novel technique critical for heterogeneous sequence lengths, contributing 15 points by dynamically adjusting the bias-variance tradeoff.

4

VAPO achieves 60.4 on AIME 2024 (avg@32) within ~5,000 steps — highly training-efficient with zero crashes across multiple independent runs.

5

Implications: precise token-level credit assignment via value models may be essential for scaling reasoning to harder problems requiring longer chains of thought.